Learning Criticality with a Temperature-Conditioned GFlowNet

Independent Study in Computational Statistical Physics | CPU-validated research artifact | August 2026

Abstract

A temperature-conditioned GFlowNet is trained to sample the periodic two-dimensional Ising model and to expose criticality through its learned normalization. Exact enumeration at L=4 and seeded Metropolis chains through L=12 gate every result. Fixed-temperature empirical KL is below 0.016 nats; a single conditioned model reaches L=4 exact KL below 0.0033 and L=8 observable errors below 4.35%. Curvature of learned L=8 log Z predicts T_c=2.3422. Susceptibility extrapolation and Binder crossings yield an observable consensus T_c=2.2599, compared with exact 2.2692. The study also identifies nonphysical boundary curvature in differentiated learned log Z, finite-size bias, and regimes where MCMC remains preferable.

Keywords: GFlowNet; Ising model; trajectory balance; partition function; critical temperature; finite-size scaling

1. Introduction

The two-dimensional zero-field Ising model is a compact test of generative sampling near a phase transition. Below criticality it has two symmetry-related ordered modes; around criticality it develops long correlations; and its partition function is normally the hard normalization in the Boltzmann law. This study asks whether a small generative flow network (GFlowNet), trained on one Mac CPU, can reproduce the distribution and predict critical temperature from its learned structure rather than merely copying a known answer.

The study has three safeguards against impressive-looking but incorrect results. First, all 65,536 states of the L=4 lattice are exactly enumerated. Second, seeded Metropolis-Hastings sampling provides an independent baseline at every tested temperature. Third, milestone validators enforce numerical thresholds and write immutable JSON provenance. The exact thermodynamic-limit value, T_c=2/ln(1+sqrt(2))=2.269185, is reserved for reference and is never supplied to training [4].

The central contribution is two independent predictions. Curvature of the L=8 model's learned log partition function gives T_c=2.3422. Finite-size susceptibility and Binder statistics from generated samples give an observable consensus T_c=2.2599. Their disagreement is scientifically useful: it exposes the different finite-size and numerical-derivative errors of the two routes.

2. Model and Methods

2.1 Ising target and observables

Spins s_i are -1 or +1 on an L by L periodic square lattice with nearest-neighbor coupling J=1 and zero field. Counting every bond once, the Hamiltonian and Boltzmann law are

$$E(x) = - sum\langle ij\rangle\ s_i s_j, \quad p_T(x) = exp[-E(x)/T] / Z(T).$$

With N=L^2 and M=sum$_i$ s_i, the analysis records energy per site, absolute magnetization, connected absolute-magnetization susceptibility, specific heat, and Binder cumulant:

$$chi = beta\ [\langle M^2\rangle - \langle|M|\rangle^2] / N, \quad U_4 = 1 - \langle M^4\rangle / (3\langle M^2\rangle^2).$$

The absolute-magnetization convention prevents finite-volume tunneling between the two ordered signs from appearing as a spurious susceptibility. Exact L=4 values use stable log-sum-exp evaluation. Metropolis uses checkerboard single-spin sweeps with acceptance min[1, exp(-Delta E/T)], mixed starts, burn-in, thinning, and explicit independent seeds.

2.2 Raster-order GFlowNet

The source is an unassigned lattice. At raster step t, the forward policy assigns -1 or +1 to site t. The terminal reward is R_T(x)=exp[-E(x)/T]. Raster order creates one trajectory per terminal and one parent per non-source state, so backward transition probabilities equal one. General trajectory balance [2] reduces to

$$delta(x,T) = log\ Z(T) + sum_t\ log\ P_F(s_{t+1}|s_t,T) + E(x)/T.$$

Training minimizes the mean squared residual. When balance holds for every complete trajectory, policy normalization forces P_F(x|T)=R_T(x)/Z(T). Thus log Z is learned rather than supplied by the oracle. This is the normalized discrete-model use of GFlowNets described in Refs. [1-3].

2.3 Temperature conditioning and symmetry

Fixed-temperature L=4 models use two 128-unit hidden layers. The conditioned models take beta=1/T and use a small separate network for log Z(beta). A masked autoregressive MLP returns every site conditional in one training pass while blocking present and future spins. The L=8 policy uses two 256-unit hidden layers, the CPU ceiling, plus spin and beta-times-spin features.

Zero-field symmetry is imposed exactly by replacing a raw logit g(s,beta) with [g(s,beta)-g(-s,beta)]/2. Consequently q(x|beta)=q(-x|beta), the first spin is unbiased, and neither ordered sign is structurally favored. Training batches combine uniform, current-policy, noisy ordered, and block-domain configurations, paired with global flips. This off-policy mixture preserves full support while exposing low-energy and domain-wall sectors.

3. Validation Results

3.1 Physics core and fixed-temperature models

The final test suite contains 20 passing checks covering energy signs, wrapped bonds, exact enumeration, seeded sampling, autoregressive normalization and causality, spin symmetry, and numerical peak estimators. Table 1 shows the worst L=4 Metropolis deviations over the M1 temperature grid.

Metric	Worst error	Gate
Energy/site	0.1003%	1%
|m|	0.1019%	1%
Susceptibility	2.7492%	5%
Specific heat	1.2070%	5%

Table 1. Exact-oracle validation. Source: M1 JSON.

At fixed temperature, two million empirical terminals per model give KL(exact||empirical)=0.014762 at T=3 and 0.015377 at T=2. Exact enumerated model KL values are 0.000511 and 0.002200. Learned log Z errors are 0.0055% and 0.5473%, respectively (M2). The empirical KL uses a disclosed Jeffreys half-count so physically nonzero rare states do not receive literal zero histogram probability.

3.2 One conditioned model

The single conditioned L=4 model reaches exact KL 0.003255, 0.001690, and 0.000308 at T=1.8, 2.269185, and 3.0. Its maximum log Z value error over a 16-point grid is 0.1053%. Table 2 compares 200,000 generated L=8 states with 256,000 fresh Metropolis states at each temperature.

T	E error	|m| error	chi error
1.8	0.0034%	0.0281%	4.3499%
2.269	0.2359%	0.4480%	2.7736%
3.0	0.2462%	0.3822%	0.6978%

Table 2. Conditioned L=8 relative errors. Source: M3 JSON.

At $T=1.8$ the generated fraction $P(m>0)=0.498825$, meeting the 0.50 +/- 0.05 mode criterion. At higher T the raw positive fraction falls slightly because exactly zero magnetization becomes more common; conditional on nonzero magnetization the signs remain balanced.

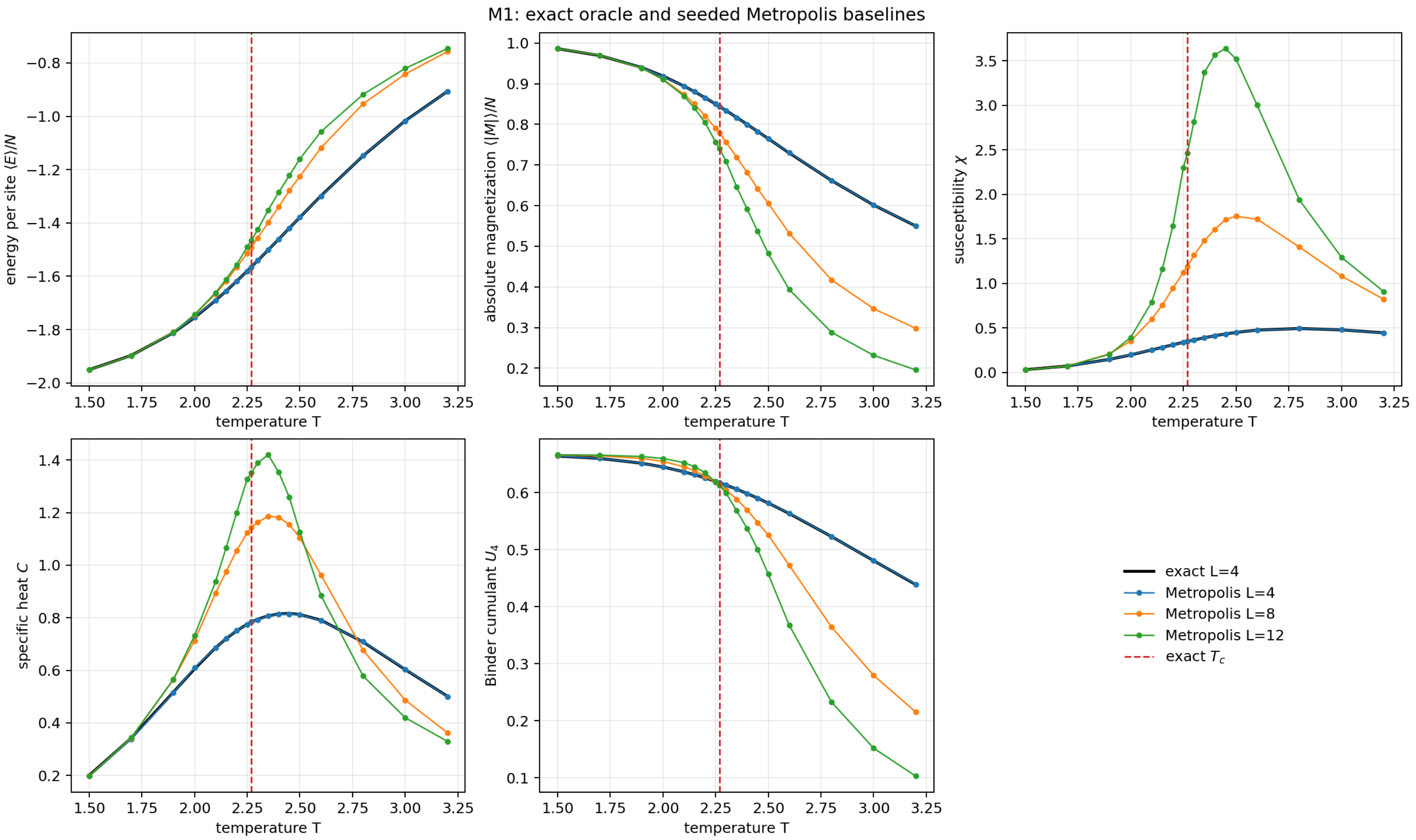

Figure 1. Exact L=4 observables and seeded Metropolis baselines at L=4, 8, and 12. Every temperature-axis panel marks the exact thermodynamic-limit critical temperature. Source: M1.

`Artifact: results/m1_observables_20260818T002648-0400.png`

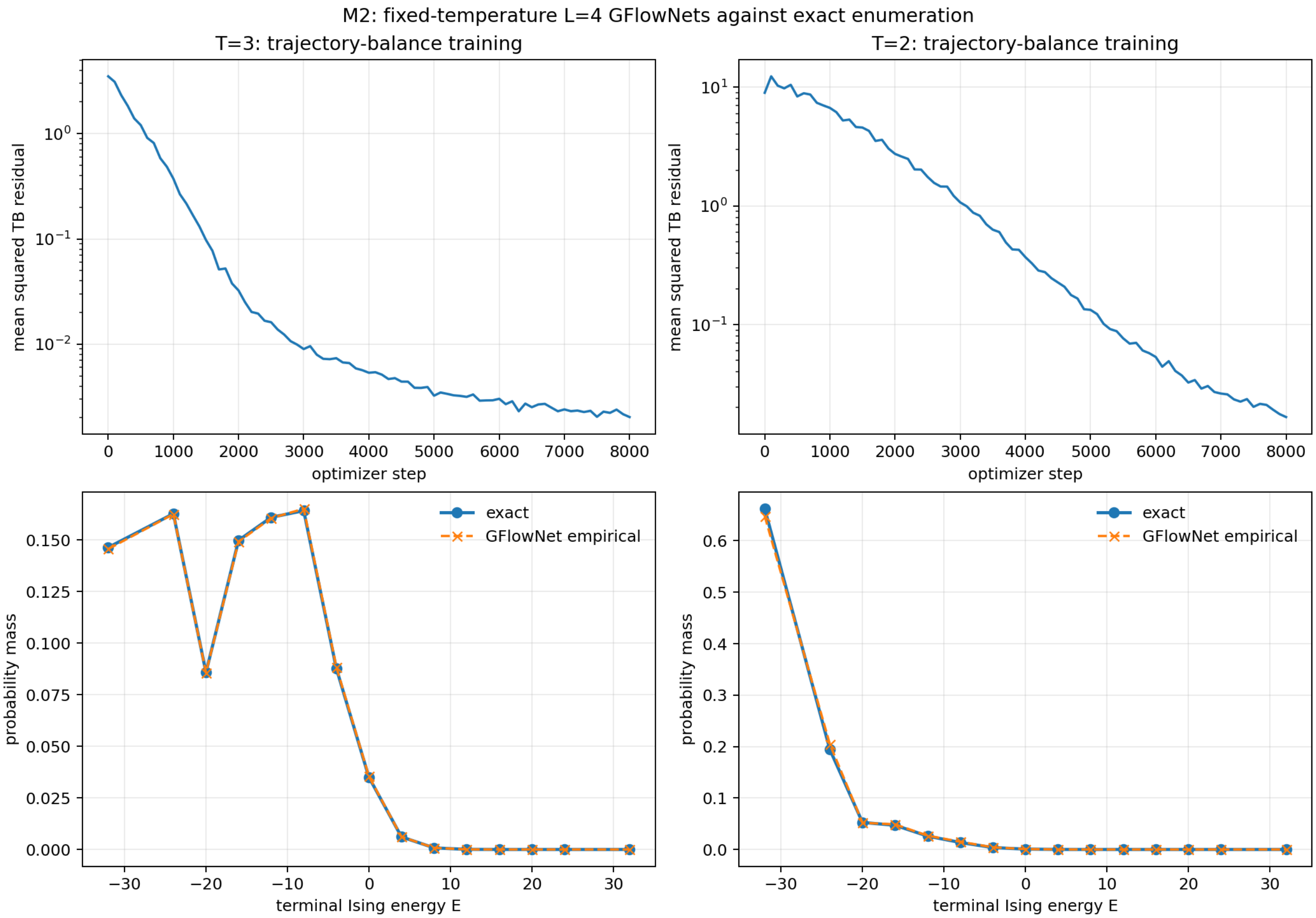

Figure 2. Fixed-temperature trajectory-balance convergence and energy-level probability mass. Empirical model mass overlays exact enumeration at both temperatures. Source: M2.

`Artifact: results/m2_fixed_temperature_20260818T012603-0400.png`

4. Predicting Critical Temperature

4.1 Learned partition-function route

The partition function supplies thermodynamics before terminal sampling. In beta coordinates, U=-d log Z/d beta and c=beta^2 d^{2}log Z/d beta^2/N. M4 evaluates log Z on 256 beta points, fits a degree-10 Chebyshev polynomial, differentiates it analytically, and searches for the interior heat-capacity maximum.

The non-negotiable calibration succeeds: direct exact L=4 observables peak at T=2.438950, while the identical exact-log-Z pipeline returns 2.439257, a 0.0126% location error. Learned log Z gives T=2.281360 at L=4 and T=2.342234 at L=8. The larger-lattice primary prediction is

$$T_c^{(\log Z)} = 2.3422 \quad (+3.22\%\ \text{versus exact}).$$

The value lies in the predeclared finite-size window [2.1,2.5]. However, the learned differentiated curves become negative near the T=1.5 boundary. A canonical Ising heat capacity cannot be negative: this is a curvature and boundary artifact, not a physical discovery. The calibrated interior maximum is retained, and the failure is a central limitation of learned-normalization derivatives.

4.2 Generated-observable route

One hundred thousand new GFlowNet samples at each of 20 temperatures supply L=4 and L=8 observables; 144,000 new Metropolis samples per temperature supply L=12. Local quadratic fits place susceptibility maxima at 2.812028, 2.552567, and 2.447225 for L=4, 8, and 12. Fitting $T_{chi}(L)=T_c+a/L$ gives

$$T_c^{(chi)} = 2.273526, \quad R^2 = 0.99825.$$

Linear Binder crossings are 2.242534 for L4/L8 and 2.249908 for L8/L12, with mean 2.246221. Giving the susceptibility family and the mean Binder family equal weight defines the declared observable consensus

$$T_c^{(obs)} = 2.2599 \quad (-0.41\%\ \text{versus exact}).$$

The equal weighting is an explicit summary convention rather than a tuned fit. The component estimates remain visible: susceptibility is 0.19% high, the Binder mean is 1.01% low, and the exact value is 2.269185.

4.3 Trajectory signatures

The L=8 mean Bernoulli action entropy rises from 0.04587 nats at T=1.5 through 0.28136 at exact criticality to 0.56129 at T=3.2. This smooth increase reflects the change from ordered, predictable later actions to disordered choices. At exact criticality P(m>0|m!=0)=0.50014. These are exploratory structural signatures; no third critical temperature is extracted from them.

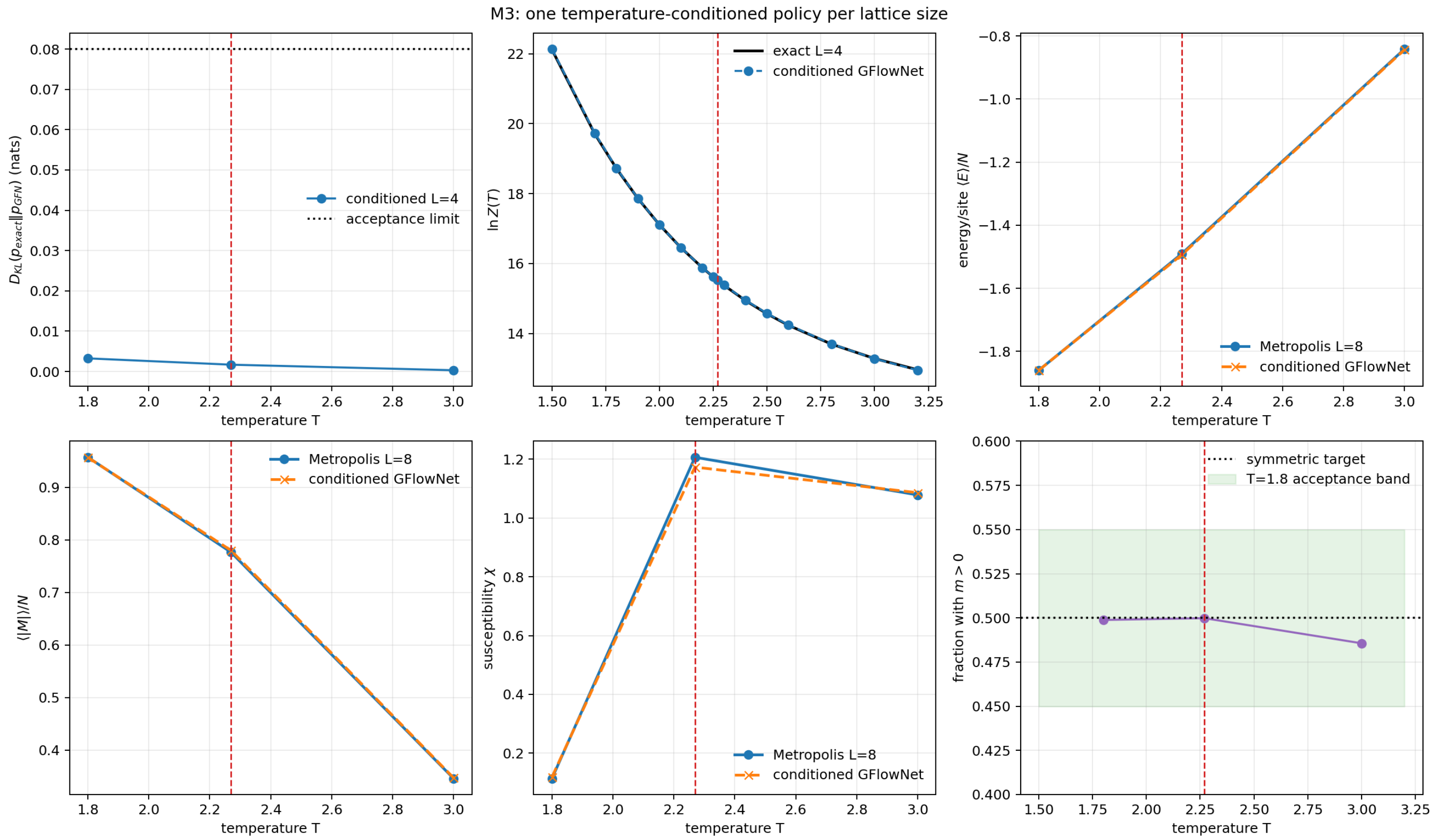

Figure 3. Validation of one temperature-conditioned policy per lattice size: exact L=4 KL and log Z, L=8 observables against Metropolis, and terminal-mode balance. Source: M3.

`Artifact: results/m3_conditioned_summary_20260818T015650-0400.png`

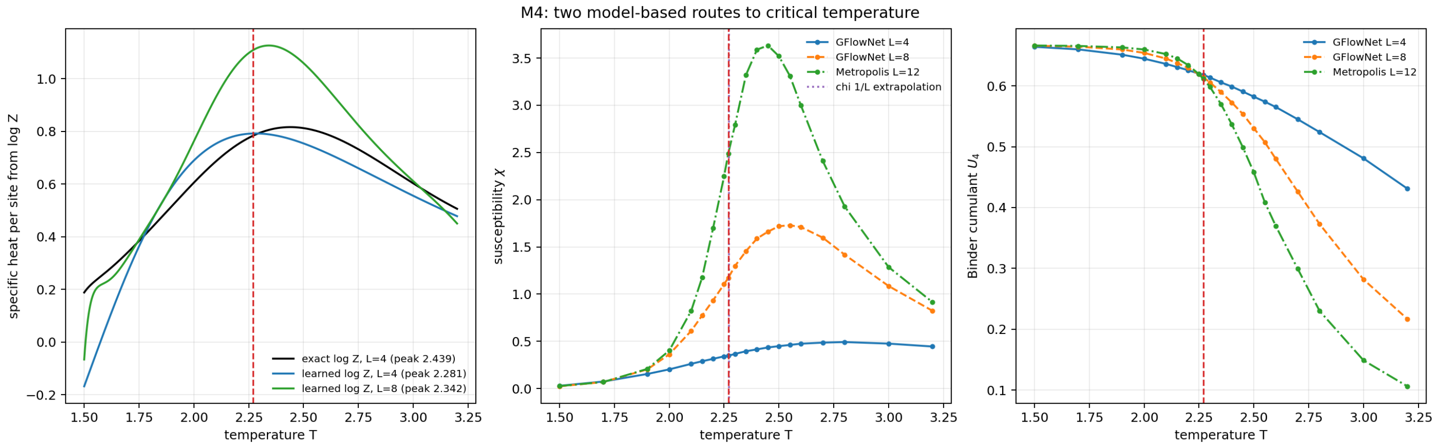

Figure 4. The required M4 summary: heat capacity from learned log Z, susceptibility, and Binder cumulant, all with the exact critical-temperature line. Source: M4.

`Artifact: results/m4_tc_summary_20260818T021212-0400.png`

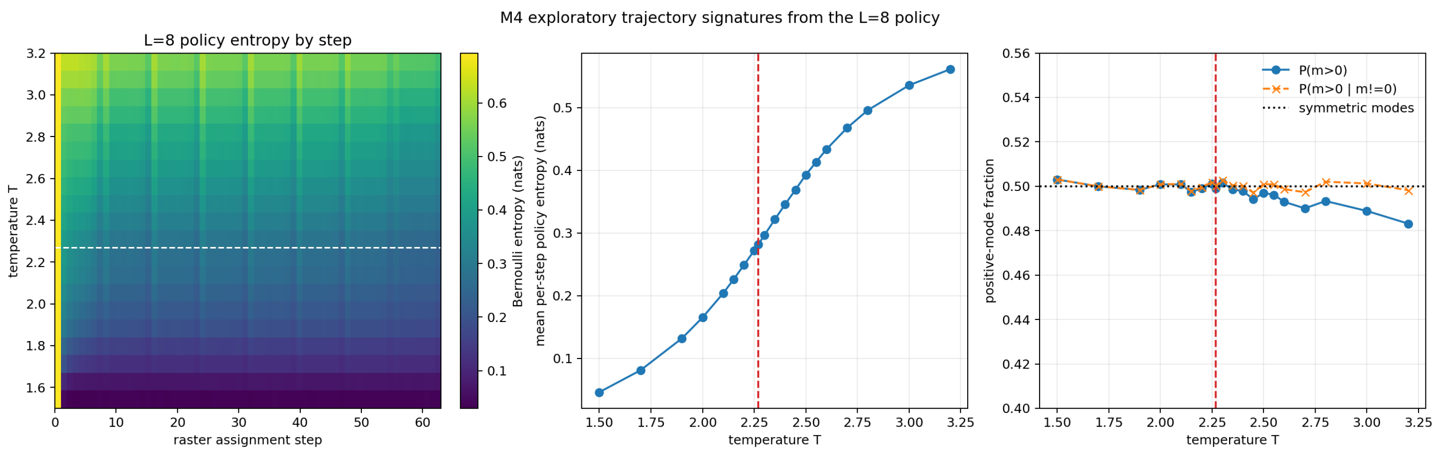

Figure 5. Exploratory L=8 policy entropy by raster step and temperature, mean entropy, and positive-mode balance. Source: M4.

`Artifact: results/m4_trajectory_signatures_20260818T021212-0400.png`

5. Discussion

The strongest evidence is the agreement of several independent objects: exact L=4 distributions, learned log Z values, L=8 generated observables, L=12 Metropolis trends, and two finite-size critical estimators. The GFlowNet does more than match a mean. Its normalized autoregressive distribution covers both ordered signs and its learned normalization contains enough curvature to locate an interior heat-capacity peak.

Finite size remains the dominant physical limitation. L=4 and L=8 are far from the thermodynamic limit; their rounded susceptibility peaks shift substantially. The high R-squared of the three-point 1/L line only describes those three points and is not asymptotic evidence. Repeated training seeds and bootstrap confidence intervals are also absent, so displayed digits identify this reproducible run rather than universal precision.

Derivative sensitivity is the principal model limitation. Sub-percent errors in log Z values can become visible curvature errors after two derivatives. A future model should parameterize log Z as a convex function of beta, because its second derivative is Var(E) and must be nonnegative. Larger locality-aware policies and repeated-seed uncertainty estimates are natural follow-ups.

MCMC still wins for a one-off trusted estimate on a modest lattice: it needs no neural training, its transition rule is transparent, and it reached L=12 under the same CPU budget. The GFlowNet wins after amortization when many independent samples across temperature are required, when explicit two-mode generation matters, or when learned log Z is itself the object of study. The methods are complementary rather than interchangeable.

6. Conclusion

A compact CPU GFlowNet learned the periodic 2D Ising distribution to exact-oracle accuracy at L=4, generalized across temperature, and reproduced L=8 Metropolis observables within the declared gates. It predicted critical temperature as 2.3422 from learned partition-function curvature and 2.2599 from generated-observable consensus, compared with exact 2.2692. The observable estimate is closer; the log-Z estimate is the more distinctively model-based result and reveals the numerical fragility of thermodynamic derivatives. Small, verified lattices support these claims; larger-system accuracy is left open.

Artifact Provenance

Every quantitative claim above is regenerated by an acceptance-gated validator. Checkpoints are SHA-256 hashed in their JSON records. The report and this PDF introduce no uncomputed table entries.

ID	Role	Primary artifact
M1	Exact + MCMC	results/m1_metrics_20260818T002648-0400.json
M2	Fixed-T GFN	results/m2_metrics_20260818T012603-0400.json
M3	Conditioned GFN	results/m3_metrics_20260818T015650-0400.json
M4	Criticality	results/m4_metrics_20260818T021212-0400.json
M5	Report audit	results/m5_metrics_20260818T022241-0400.json

Table 3. Machine-readable provenance map.

References

[1] Y. Bengio et al., “GFlowNet Foundations,” arXiv:2111.09266 (2021).

[2] N. Malkin et al., “Trajectory Balance: Improved Credit Assignment in GFlowNets,” arXiv:2201.13259 (2022).

[3] D. Zhang et al., “Generative Flow Networks for Discrete Probabilistic Modeling,” arXiv:2202.01361 (2022).

[4] L. Onsager, “Crystal Statistics. I. A Two-Dimensional Model with an Order-Disorder Transition,” Physical Review 65, 117-149 (1944).